Shuchir Harsha

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EDUCATION

University of California at San Diego — La Jolla, CA

Bachelor of Science in Computer Science and Mathematics — *Expected Graduation: June 2027*

- Courses: Algorithms, Advanced Data Structures, Discrete Mathematics & Graph Theory, Computer Organization & Assembly, Digital Design, Probability & Statistics, Linear Algebra, Numerical Analysis
- Leadership & Clubs: Eagle Scout (led planning for a 60-person troop), Triton Robotics Embedded Team, SWE & CE Club

EXPERIENCE

Hewlett Packard Enterprise (Juniper Mist AI) — Sunnyvale, CA

Cloud Engineering Intern, Cloud Data Platform — *Jun 2026 - Present*

- Built application QoS statistics processing into a production telemetry pipeline on **Kafka, Apache Storm, and Elasticsearch**, computing **ACK RTT, TCP retransmissions, congestion window, throughput, and flow counts** per client and per application.
- Wrote an Apache **Storm bolt** that joins per-application traffic statistics with wireless client and radio telemetry into a single correlated record, so a slow application can be traced back to the wireless link, the wired network, or the WAN.
- Built a **Java 11 Flink sink** that bulk indexes into Elasticsearch with checkpointing, retry handling, and **idempotent writes**, replacing a heavier client-based path. Covered it with integration tests against a mock HTTP server and deployed it to **AWS staging**.
- Added **root-cause scoring** to the QoS detector, labeling each degraded app with the metric driving the problem.
- Added **REST endpoints** aggregating Elasticsearch QoS data into per-app performance metrics for App Insights dashboards.
- Split the **Kafka to ORC archival path** into per-topic writers and renamed topics across AWS/GCP staging, coordinating **4 services**.
- Wrote **Python and Bash tooling** cross-checking Kafka, Elasticsearch, and S3 records against raw packet captures before releases.

NETGEAR — San Jose, CA

Software Engineer Intern — *Jun 2025 - Sep 2025*

- Built **Python backend services** turning plain-English requests into validated NETGEAR Insight cloud API calls for diagnostics.
- Wrote async REST and JSON-RPC clients with **asyncio and httpx** to query a fleet of devices concurrently, not one at a time.
- Handled device communication with dynamic API discovery, session-based auth, and TLS, keeping credentials out of tooling.
- Ran **retrieval-augmented generation** on **local LLMs** so diagnostics stayed on-premises and customer data never left.

NeuroLeap — San Jose, CA

Embedded Software Intern — *Apr 2024 - Oct 2024*

- Modified **Android 12** to run **Debian 11** via **proot and VNC** so the team could build gateway software with standard Linux tooling.
- Wrote **UART and I2C** communication code for flashing microcontrollers and driving peripherals, including 7-segment displays.
- Set up embedded Linux build and test workflows for **Arduino and Raspberry Pi** to shorten firmware iteration.

PROJECTS

Object Detecting Camera | Python, TensorFlow Lite, Embedded Linux

- Real-time object detection on constrained embedded hardware at **roughly 99% accuracy** after quantizing the model to fit.

AI-Powered Network Diagnostics Agent | Python, LLM tooling, REST

- Agent backend that controls network devices via chat, with a **validation layer** rejecting unsafe commands before execution.

TECHNICAL SKILLS

Languages: Java, Python, Go, C++

Frameworks & Tools: Apache Kafka, Apache Storm, Apache Flink, Elasticsearch, AWS, GCP, Embedded Linux, REST, JSON-RPC, Git